ResStock demo, market results (hyperbolic)

Local energy sharing AMM on measured building data and real day-ahead prices

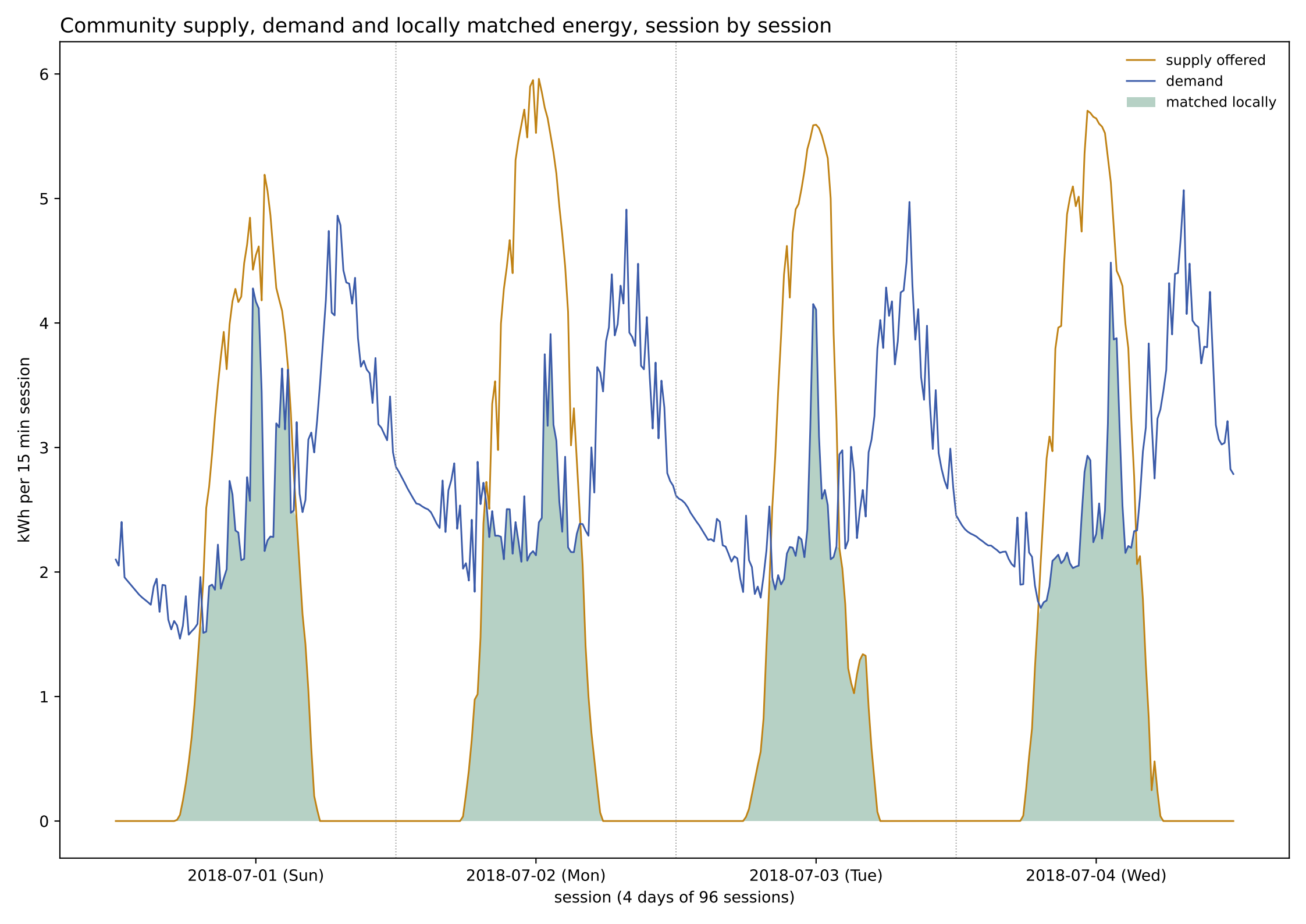
Community	10 prosumers (5 with rooftop PV, 5 without), single Massachusetts county, ResStock amy2018 release 2
Period	4 trading days, 2018-07-01 to 2018-07-04, 96 sessions of 15 minutes per day
Prices	ISO New England day-ahead LMP, zone WCMA, retail = wholesale + 0.12 USD/kWh delivery adder
Status quo	every kWh traded with the grid alone: exports paid low (wholesale), imports paid high (retail)

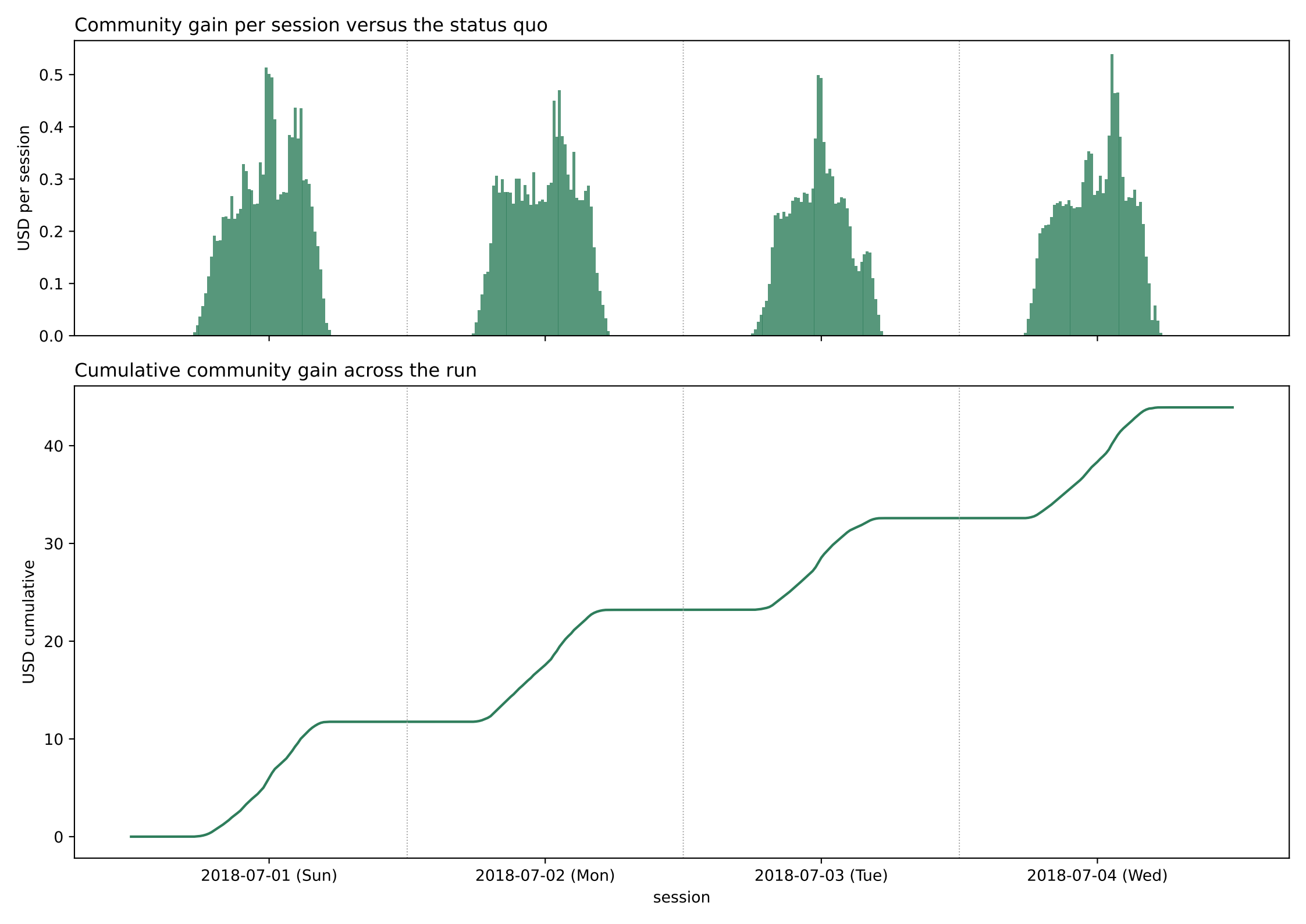
Headline results

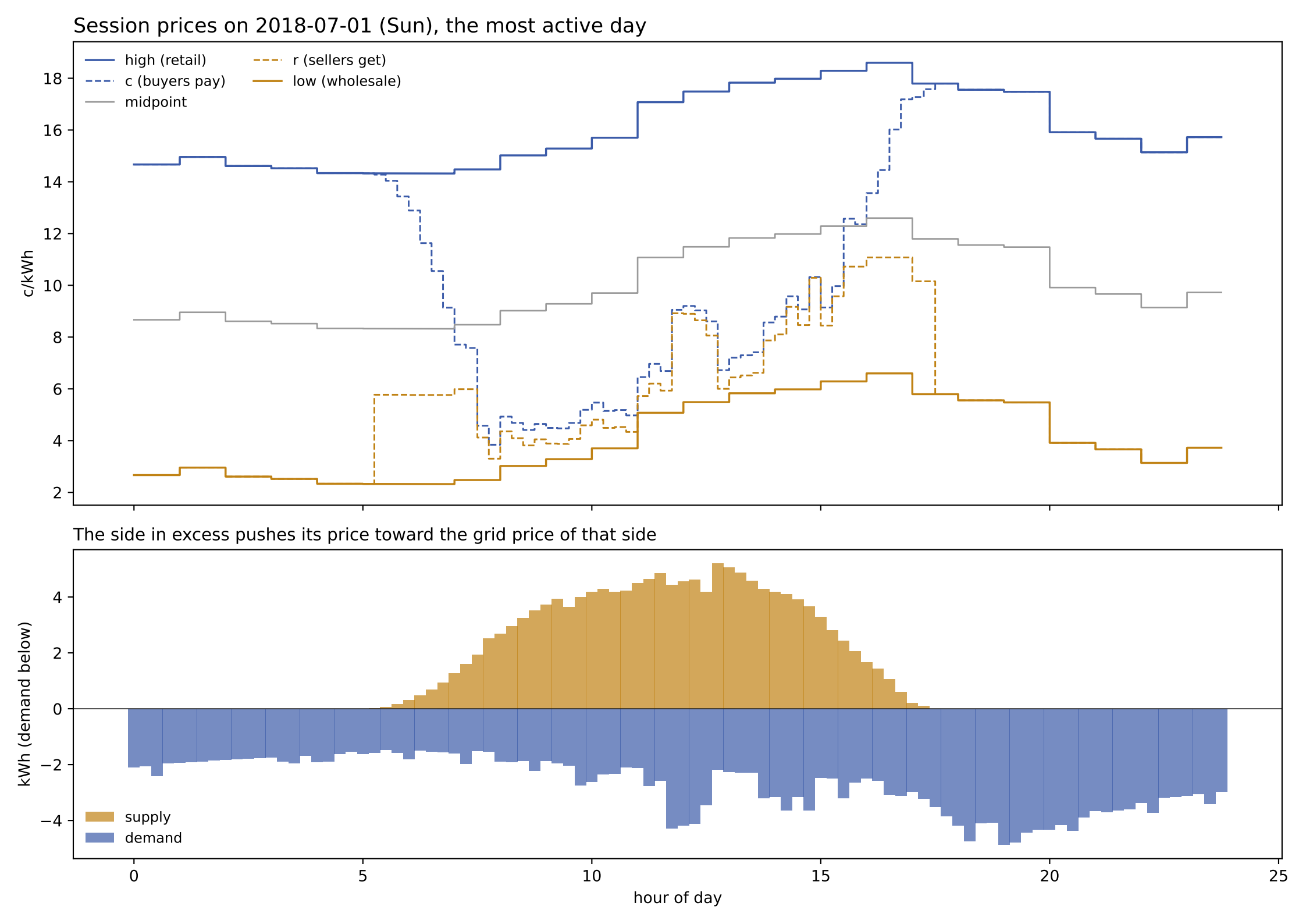
- Community gain over 4 days: 43.92 USD (10.98 USD/day), entirely the delivery spread earned on locally matched energy
- Locally matched energy: 366.0 kWh of 1 064.6 kWh demanded, 34.4 % of demand served locally (61.6 % of the surplus absorbed locally)
- Sellers received on average 7.07 c/kWh against 5.57 wholesale; buyers paid 13.35 c/kWh against 16.64 retail
- Spread high - low = the 12 c/kWh delivery adder in every session, so the gain is 0.12 USD per matched kWh and matching volume is the only lever

Daily market summary

	offered kWh	demanded kWh	matched kWh	covered %	absorbed %	gain USD
2018-07-01 (Sun)	141.49	259.83	97.95	37.70	69.23	11.75
2018-07-02 (Mon)	164.14	272.99	95.42	34.95	58.13	11.45
2018-07-03 (Tue)	129.07	265.02	78.20	29.51	60.58	9.38
2018-07-04 (Wed)	159.78	266.76	94.42	35.39	59.09	11.33


Community supply, demand and locally matched energy, session by session
supply offered
demand
matched locally
kWh per 15 min session
0
1
2
3
4
5
6
2018-07-01 (Sun)
2018-07-02 (Mon)
2018-07-03 (Tue)
2018-07-04 (Wed)
session (4 days of 96 sessions)

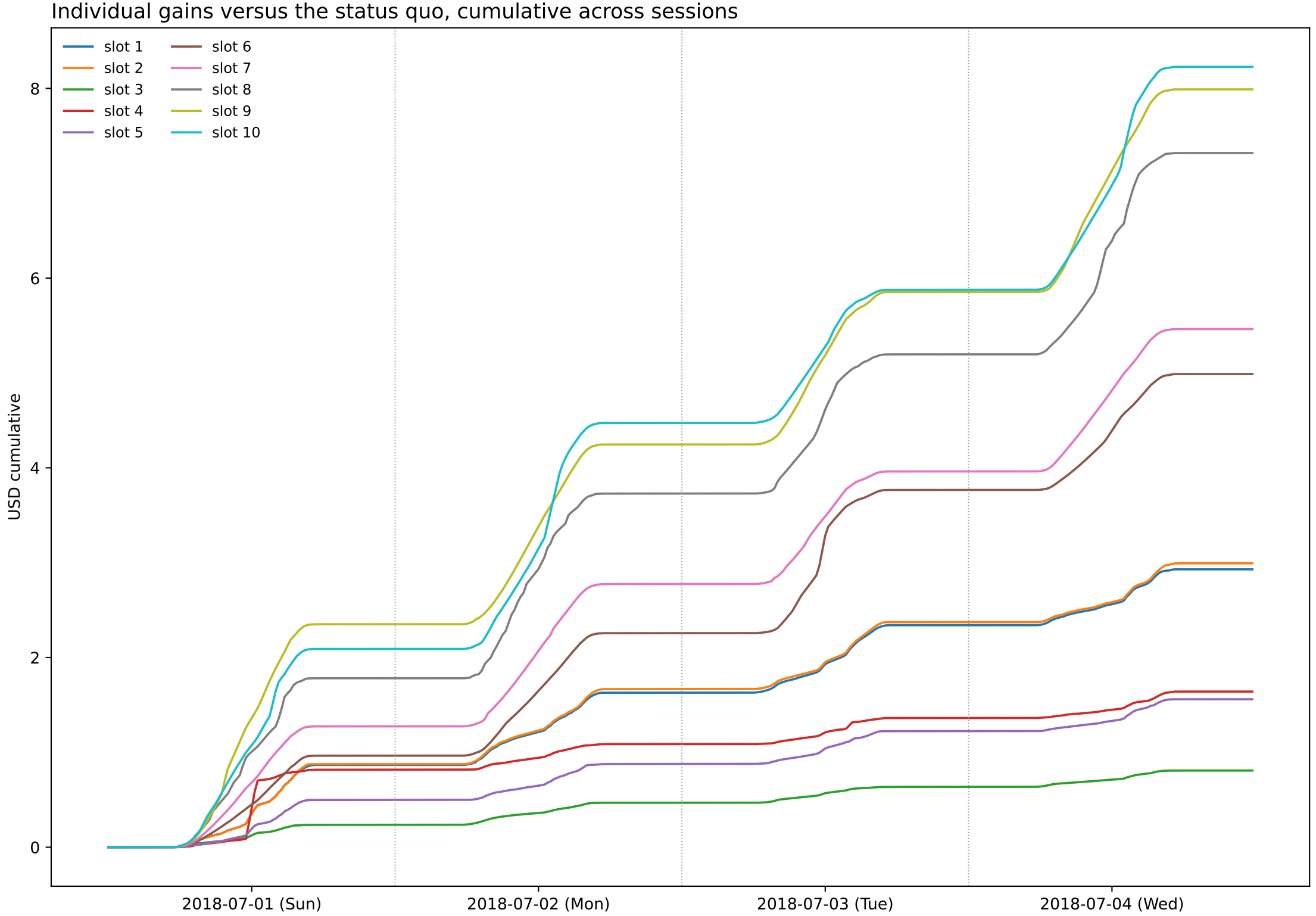

Community gain per session versus the status quo
USD per session
0.0
0.1
0.2
0.3
0.4
0.5
Cumulative community gain across the run
USD cumulative
0
10
20
30
40
2018-07-01 (Sun)
2018-07-02 (Mon)
2018-07-03 (Tue)
2018-07-04 (Wed)
session

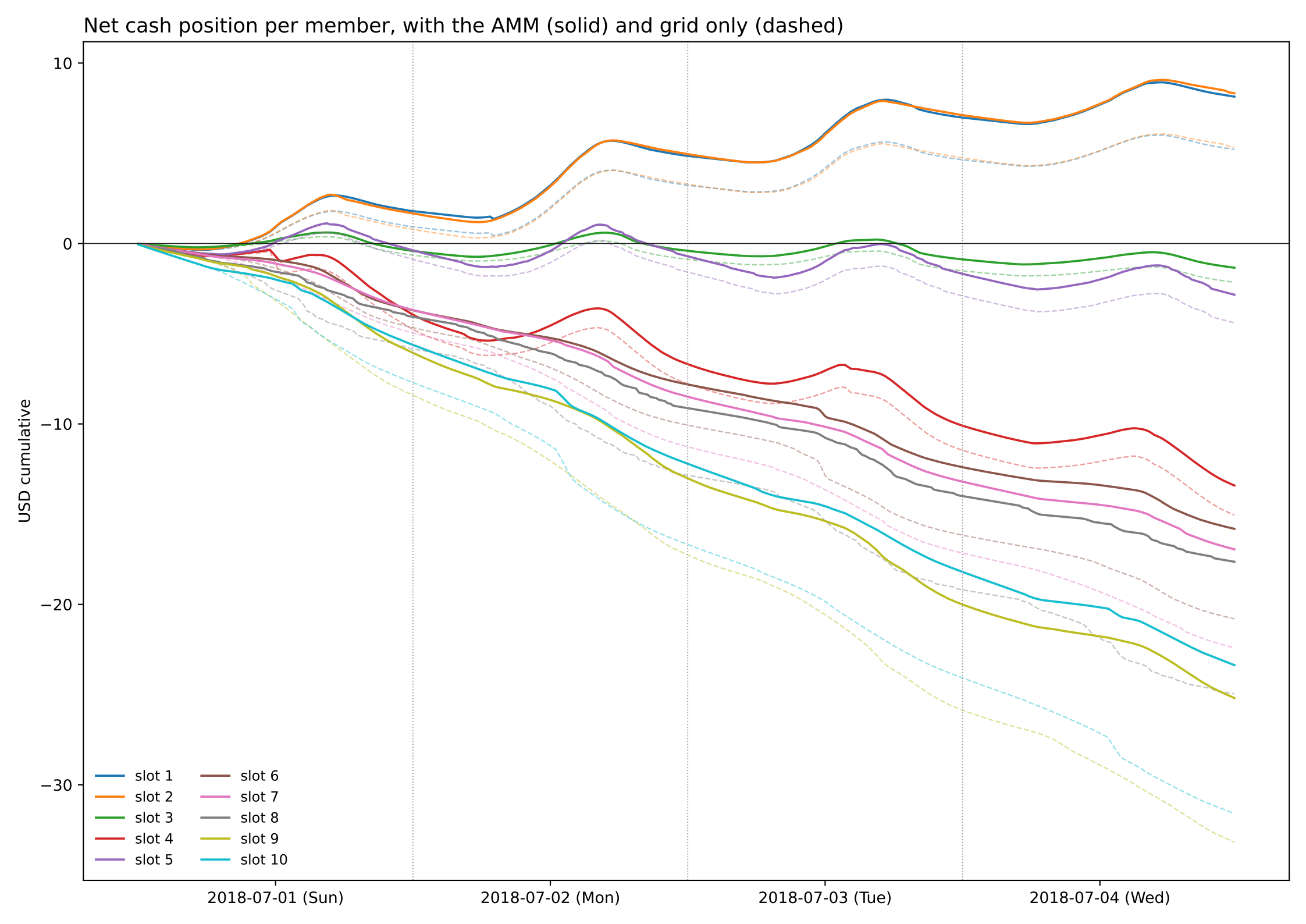

Session prices on 2018-07-01 (Sun), the most active day
high (retail)
c (buyers pay)
midpoint
r (sellers get)
low (wholesale)
c/kWh
18
16
14
12
10
8
6
4
2
The side in excess pushes its price toward the grid price of that side
kWh (demand below)
4
2
0
−2
−4
supply
demand
0
5
10
15
20
25
hour of day

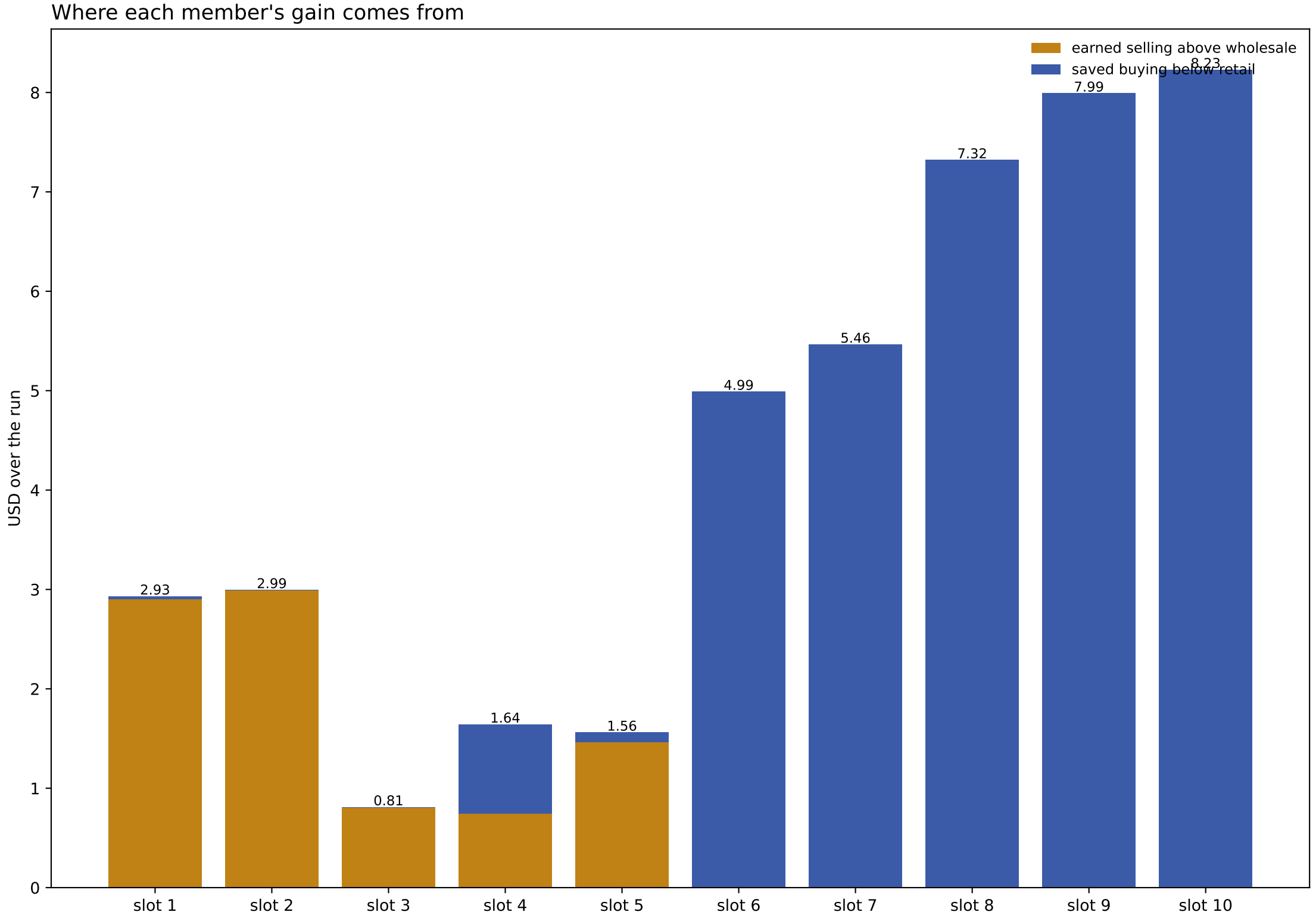
Individual results over the whole run

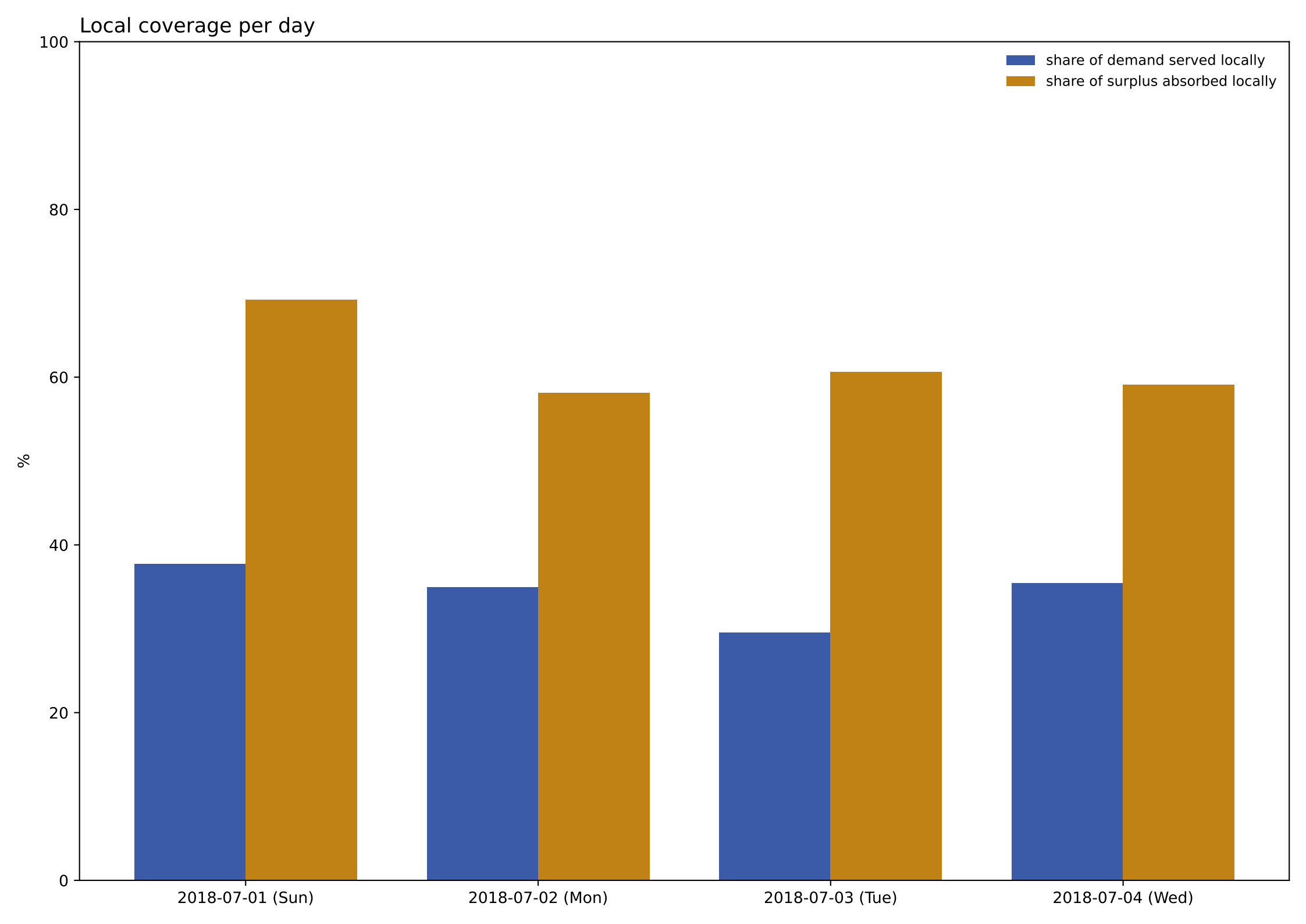
Cash flows in USD, negative = net payment. Gain = with AMM - grid only; it is non negative for every member in every session.

	sold kWh	bought kWh	grid only USD	with AMM USD	gain USD	gain c/kWh traded	share of gain %
slot 1	184.40	32.10	5.21	8.14	2.93	1.353	6.7
slot 2	184.96	31.99	5.33	8.32	2.99	1.380	6.8
slot 3	55.25	31.63	-2.15	-1.34	0.81	0.930	1.8
slot 4	67.28	114.88	-15.05	-13.41	1.64	0.901	3.7
slot 5	102.60	64.75	-4.40	-2.84	1.56	0.932	3.6
slot 6	0.00	122.28	-20.80	-15.81	4.99	4.080	11.4
slot 7	0.00	132.90	-22.41	-16.95	5.46	4.110	12.4
slot 8	0.00	147.52	-24.96	-17.64	7.32	4.961	16.7
slot 9	0.00	198.18	-33.18	-25.19	7.99	4.032	18.2
slot 10	0.00	188.38	-31.59	-23.37	8.23	4.367	18.7


Individual gains versus the status quo, cumulative across sessions
slot 1
slot 2
slot 3
slot 4
slot 5
slot 6
slot 7
slot 8
slot 9
slot 10
USD cumulative
0
2
4
6
8
2018-07-01 (Sun)
2018-07-02 (Mon)
2018-07-03 (Tue)
2018-07-04 (Wed)


Net cash position per member, with the AMM (solid) and grid only (dashed)
10
0
−10
−20
−30
USD cumulative
slot 1
slot 2
slot 3
slot 4
slot 5
slot 6
slot 7
slot 8
slot 9
slot 10
2018-07-01 (Sun)
2018-07-02 (Mon)
2018-07-03 (Tue)
2018-07-04 (Wed)


Where each member's gain comes from
earned selling above wholesale
saved buying below retail
USD over the run
0
1
2
3
4
5
6
7
8
2.93
2.99
0.81
1.64
1.56
4.99
5.46
7.32
7.99
8.23
slot 1
slot 2
slot 3
slot 4
slot 5
slot 6
slot 7
slot 8
slot 9
slot 10


Local coverage per day
share of demand served locally
share of surplus absorbed locally
%
100
80
60
40
20
0
2018-07-01 (Sun)
2018-07-02 (Mon)
2018-07-03 (Tue)
2018-07-04 (Wed)